Problem Analysis

Project Title: Student Record and Academic Management System – Data Organization using Python Collections

Problem Statement

The objective is to develop a Student Record and Academic Management System using Python. The system stores and manages student academic information such as roll number, name, department, registration number, date of birth, subjects, marks, attendance, email, phone number and address.

Objectives

• Understand Python built-in collections.

• Add, search, update and delete student records.

• Calculate average marks and find the highest scorer.

• List students according to department and count students.

• Demonstrate strings, lists, tuples, sets and dictionaries.

• Integrate multiple collections into one complete application.

Inputs

• Roll number, name, department, registration number, date of birth, subjects, marks, attendance, email, phone, address and menu choice.

Outputs

• Student details, marks, attendance, average, highest scorer, department-wise students, total count, unique departments and unique subjects.